Now use (1) in the foregoing expressions for v_r and v_θ. By comparing your results with the expressions for u_r and u_θ, deduce the Cauchy-Riemann equations in polar coordinates given in (10).

39. Suppose the function $f(z) = u(r, \theta) + iv(r, \theta)$ is differentiable at a point z whose polar coordinates are (r, θ). Solve the two equations in (12) for u_x and then solve the two equations in (13) for v_x. Then show that the derivative of f at (r, θ) is

$$f'(z) = (\cos\theta - i\sin\theta)\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right) = e^{-i\theta}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right).$$

40. Consider the function

$$f(z) = \begin{cases} 0, & z = 0 \\ \dfrac{z^5}{|z^4|}, & z \neq 0. \end{cases}$$

(**a**) Express f in the form $f(z) = u(x, y) + iv(x, y)$,

(**b**) Show that f is not differentiable at the origin.

(**c**) Show that the Cauchy-Riemann equations are satisfied at the origin. [*Hint*: Use the limit definitions of the partial derivatives $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, and $\partial v/\partial y$ at $(0, 0)$.]

3.4 Harmonic Functions

INTRODUCTION In Section 5.5 we shall see that when a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic at a point z, then *all* the derivatives of f: $f'(z)$, $f''(z)$, $f'''(z)$, ... are also analytic at z. As a consequence of this remarkable fact, we can conclude that *all* partial derivatives of the real functions $u(x, y)$ and $v(x, y)$ are continuous at z. From the continuity of the partial derivatives we then know that the second-order mixed partial derivatives are equal. This last fact, coupled with the Cauchy-Riemann equations, will be used in this section to demonstrate that there is a connection between the real and imaginary parts of an analytic function $f(z) = u(x, y) + iv(x, y)$ and the second-order partial differential equation

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0. \tag{1}$$

This equation, one of the most famous in applied mathematics, is known as **Laplace's equation** in two variables. The sum $\dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial^2\phi}{\partial y^2}$ of the two second-order partial derivatives in (1) is denoted by $\nabla^2\phi$ and is called the **Laplacian** of ϕ. Laplace's equation (1) is then abbreviated as $\nabla^2\phi = 0$.

Harmonic Functions

A solution $\phi(x, y)$ of Laplace's equation (1) in a domain D of the plane is given a special name.

DEFINITION 3.4.1 Harmonic Functions

A real-valued function ϕ of two real variables x and y that has continuous first and second-order partial derivatives in a domain D and satisfies Laplace's equation is said to be **harmonic** in D.

Harmonic functions are encountered in the study of temperatures and potentials. These applications are introduced in Section 3.5. In the next theorem we see that both the real and the imaginary parts of an analytic function are harmonic.

THEOREM 3.4.1 Harmonic Functions

Suppose the complex function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D. Then the functions $u(x, y)$ and $v(x, y)$ are harmonic in D.

Proof Assume $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D and that u and v have continuous second-order partial derivatives in D.* Since f is analytic, the Cauchy-Riemann equations are satisfied at every point z. Differentiating both sides of $\partial u/\partial x = \partial v/\partial y$ with respect to x and differentiating both sides of $\partial u/\partial y = -\partial v/\partial x$ with respect to y give, respectively,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}. \tag{2}$$

With the assumption of continuity, the mixed partials $\partial^2 v/\partial x \partial y$ and $\partial^2 v/\partial y \partial x$ are equal. Hence, by adding the two equations in (2) we see that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{or} \quad \nabla^2 u = 0.$$

This shows that $u(x, y)$ is harmonic.

Now differentiating both sides of $\partial u/\partial x = \partial v/\partial y$ with respect to y and differentiating both sides of $\partial u/\partial y = -\partial v/\partial x$ with respect to x, give, in turn, $\partial^2 u/\partial y \partial x = \partial^2 v/\partial y^2$ and $\partial^2 u/\partial x \partial y = -\partial^2 v/\partial^2 x$. Subtracting the last two equations yields $\nabla^2 v = 0$. ☰

EXAMPLE 1 Harmonic Functions

The function $f(z) = z^2 = x^2 - y^2 + 2xyi$ is entire. Therefore, the functions $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$ are necessarily harmonic in any domain D of the complex plane. This is also easy to verify directly:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2 = 0 \quad \text{and} \quad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 + 0 = 0.$$ ☰

Harmonic Conjugate Functions We have just shown that if a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D, then its real and imaginary parts u and v are necessarily harmonic in D. Now suppose $u(x, y)$ is a given real function that is known to be harmonic in D. If it is possible to find another real harmonic function $v(x, y)$ so that u and v satisfy the Cauchy-Riemann equations throughout the domain D, then the function $v(x, y)$ is called a **harmonic conjugate** of $u(x, y)$. By combining the functions as $u(x, y) + iv(x, y)$ we obtain a complex function that is analytic in D.

EXAMPLE 2 Harmonic Conjugate

(**a**) Verify that the function $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic in the entire complex plane.

(**b**) Find a harmonic conjugate of u.

*The continuity of the second-order partial derivatives of u and v is not part of the hypothesis of the theorem. This fact will be proved in Chapter 5.

Solution (**a**) From the partial derivatives

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial^2 u}{\partial x^2} = 6x, \quad \frac{\partial u}{\partial y} = -6xy - 5, \quad \frac{\partial^2 u}{\partial y^2} = -6x$$

we see that u satisfies Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x - 6x = 0.$$

(**b**) Since a harmonic conjugate v must satisfy the Cauchy-Riemann equations $\partial v/\partial y = \partial u/\partial x$ and $\partial v/\partial x = -\partial u/\partial y$, we must have

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2 \quad \text{and} \quad \frac{\partial v}{\partial x} = 6xy + 5. \tag{3}$$

Partial integration of the first equation in (3) with respect to the variable y gives $v(x, y) = 3x^2y - y^3 + h(x)$. The partial derivative with respect to x of this last equation is

$$\frac{\partial v}{\partial x} = 6xy + h'(x).$$

When this result is substituted into the second equation in (3) we obtain $h'(x) = 5$, and so $h(x) = 5x + C$, where C is a real constant. Therefore, the harmonic conjugate of u is $v(x, y) = 3x^2y - y^3 + 5x + C$. ☰

In Example 2, by combining u and its harmonic conjugate v as $u(x, y) + iv(x, y)$, the resulting complex function

$$f(z) = x^3 - 3xy^2 - 5y + i(3x^2y - y^3 + 5x + C)$$

is an analytic function throughout the domain D consisting, in this case, of the entire complex plane. In Example 1, since $f(z) = z^2 = x^2 - y^2 + 2xyi$ is entire, the real function $v(x, y) = 2xy$ is a harmonic conjugate of $u(x, y) = x^2 - y^2$. See Problem 22 in Exercises 3.4.

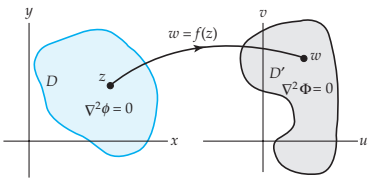


FIGURE 3.4.1 A solution of Laplace's equation in D is found by solving it in D'.

Remarks COMPARISON WITH REAL ANALYSIS

In this section we have seen if $f(z) = u(x, y) + iv(x, y)$ is an analytic function in a domain D, then both functions u and v satisfy $\nabla^2\phi = 0$ in D. There is another important connection between analytic functions and Laplace's equation. In applied mathematics it is often the case that we wish to solve Laplace's equation $\nabla^2\phi = 0$ in a domain D in the xy-plane, and for reasons that depend in a very fundamental manner on the shape of D, it simply may not be possible to determine ϕ. *But* it may be possible to devise a special analytic mapping $f(z) = u(x, y) + iv(x, y)$ or

$$u = u(x, y), \quad v = v(x, y), \tag{4}$$

from the xy-plane to the uv-plane so that D', the image of D under (4), not only has a more convenient shape but the function $\phi(x, y)$ that satisfies Laplace's equation in D also satisfies Laplace's equation in D'. We then solve Laplace's equation in D' (the solution Φ will be a function of u and v) and then return to the xy-plane and $\phi(x, y)$ by means of (4). This *invariance* of Laplace's equation under the mapping will be utilized in Chapters 4 and 7. See FIGURE 3.4.1.

EXERCISES 3.4

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1–10, (**a**) verify that the given function u is harmonic in an appropriate domain D, (**b**) find $v(x, y)$, a harmonic conjugate of u, and (**c**) form the corresponding analytic function $f(z) = u + iv$.

1. $u(x, y) = x$

2. $u(x, y) = 2x - 2xy$

3. $u(x, y) = x^2 - y^2$

4. $u(x, y) = x^3 - 3xy^2$

5. $u(x, y) = \log_e(x^2 + y^2)$

6. $u(x, y) = \cos x \cosh y$

7. $u(x, y) = e^x(x \cos y - y \sin y)$

8. $u(x, y) = -e^{-x} \sin y$

9. $u(x, y) = \dfrac{x}{x^2 + y^2}$

10. $u(x, y) = \arctan(-y/x)$

In Problems 11 and 12, verify that the given function u is harmonic in an appropriate domain D. Find a harmonic conjugate v and find an analytic function $f(z) = u + iv$ satisfying the indicated condition.

11. $u(x, y) = xy + x + 2y - 5$; $f(2i) = -1 + 5i$

12. $u(x, y) = 4xy^3 - 4x^3y + x$; $f(1 + i) = 1 + 4i$

13. (**a**) Show that $v(x, y) = \dfrac{x}{x^2 + y^2}$ is harmonic in a domain D not containing the origin.

(**b**) Find a function $f(z) = u(x, y) + iv(x, y)$ that is analytic in domain D.

(**c**) Express the function f found in part (b) in terms of the symbol z.

14. Suppose $f(z) = u(r, \theta) + iv(r, \theta)$ is analytic in a domain D not containing the origin. Use the Cauchy-Riemann equations (10) of Section 3.3 in the form $ru_r = v_\theta$ and $rv_r = -u_\theta$ to show that $u(r, \theta)$ satisfies Laplace's equation in polar coordinates:

$$r^2 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0. \tag{5}$$

In Problems 15 and 16, (**a**) use (5) to verify that the given function u is harmonic in a domain D not containing the origin, (**b**) use equations (10) of Section 3.3 to find v, a harmonic conjugate of u, and (**c**) form the corresponding analytic function $f(z) = u + iv$.

15. $u(r, \theta) = r^3 \cos 3\theta$

16. $u(r, \theta) = \dfrac{10r^2 - \sin 2\theta}{r^2}$

Focus on Concepts

17. (**a**) Verify that $u(x, y) = e^{x^2 - y^2} \cos 2xy$ is harmonic in an appropriate domain D.

(**b**) Find a harmonic conjugate v and find an analytic function $f(z) = u + iv$ satisfying $f(0) = 1$. [*Hint*: When integrating, think of reversing the product rule.]

18. Suppose that $f(z)$ is an entire function, $\text{Re}(f(z)) = y^3 - 3x^2y + 2$, and $f(i) = 2 - i$. Find $f(1 + i)$.

19. Express the function f found in Problem 11 in terms of the symbol z.

20. Express the function f found in Problem 16 in terms of the symbol z.

21. (**a**) Show that $\phi(x, y, z) = \dfrac{1}{\sqrt{x^2 + y^2 + z^2}}$ is harmonic, that is, satisfies Laplace's equation $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$ in a domain D of space not containing the origin.

(**b**) Is the two-dimensional analogue of the function in part (a), $\phi(x, y) = \dfrac{1}{\sqrt{x^2 + y^2}}$, harmonic in a domain D of the plane not containing the origin?